King Abdulaziz University

Faculty of Computing & Info Tech.
Computer Science Department

Midterm-Exam (24-10-1444 / 14-05-2023)

Course: CPCS203 (Programming II)

Term: Spring (2023), Time: 80 Minutes

Student ID: --------------------**Section:** --

Student Name: --

Instructors: Mark [✓], Your Theory Instructor

Dr. Emad [ ]   Dr. Asif [ ]

Section	Max. Mark	Obtained Mark	Total Mark
I	05		
II	15		
III	05		

Part I (5 marks)

Answer the following MCQ questions:

1. Given a **String** variable named **line1**, Which statements that used to read the first line of a file named "Story" and stores it in **line1**. [0.5 marks]

a) File fileInput = new File("Story");
Scanner scan = new Scanner(fileInput);
String line1 = scan.nextLine();

b) File fileInput = new File("Story");
Scanner scan = new Scanner(fileInput);
String line1 = fileInput.nextLine();

c) File fileInput = new File(Story);
Scanner scan = new Scanner (fileInput);
String line1 = scan.nextLine();

d) File fileInput = new File("Story");
Scanner scan = new Scanner();
String line1 = scan.next();

2. What is the output of the following program? [0.5 marks]

```
import java.io.FileNotFoundException;
import java.io.PrintWriter;

public class MidExamQuestion {
    public static void main(String[] args) throws FileNotFoundException {
        java.io.File file = new java.io.File("input.txt");
        PrintWriter output = new java.io.PrintWriter(file);
        output.println("Hello \n world Java");
        output.close();
        java.util.Scanner input = new java.util.Scanner(file);
        System.out.println(input.nextLine());
    }
}
```

a) world Java
b) Hello world Java
c) Print nothing
d) Hello

3. Which of the following Regular Expressions can be used to match strings representing Saudi phone numbers that must consist of exactly 10 digits and start with “05”: [0.5 marks]

    a) 05\d{8}
    b) 05\d{10}
    c) 05.{8}
    d) {05}8

4. Which of the following statements will produce a TRUE value? [0.5 marks]

    a) "Test" == new String("Test")
    b) new String("Test") == new String("Test")
    c) "Test".equals(new String("Test"))
    d) "Test" == "test"

5. Which of the following is considered a valid constructor signature for a class named **Test**? [0.5 marks]

    a) public void Constructor()
    b) public void Test()
    c) Test()
    d) None of the above

6. Which of the following statements is considered **false** about constructors? [0.5 marks]

    a) Constructors are used to initialize objects
    b) Constructors can be invoked for an object using the access operator “.”
    c) Programmers are allowed to develop classes without implementing any constructor
    d) A class can have more than one constructor

7. **Assume S is an array of 5 Students. Which statement will return the second Student's id?** [0.5 mark]

```
public class Student {
    private int id;
    public int getId(){ return id; }
}
```

a) `int id = S[2].getId();`
b) `int id = S.getId();`
c) `int id = S[1].getId();`
d) `int id = S[1].id;`
e) `int id = (new Student()).getId();`

8. Why the following code cause an error? [1 mark]

```
public class Test extends A {
  public static void main(String[] args) {
    Test t = new Test();
    t.print();
  }
}

class A {
  String s;

  A(String s) {
    this.s = s;
  }

  public void print() {
    System.out.println(s);
  }
}
```

a) Because class Test does not have a default constructor
b) Because class A does not have a default constructor
c) Because the main is inside Test class
d) Because of the method print have no argument

9. Which keyword used to invoke a constructor or method in a superclass?

[0.5 mark]

a) `new`
b) `static`
c) `instanceof`
d) `super`
e) `this`

Part II (15 marks)

Answer the following questions by tracing the code or identifying the errors in the code:

1. What is the output of the following program? [2.0 marks]

```
import java.io.File;
import java.io.FileNotFoundException;
import java.io.PrintWriter;

public class MidExamQuestion {
    public static void main(String[] args) throws
FileNotFoundException {
        File file = new File("input.txt");
        PrintWriter output = new PrintWriter(file);

        for (int i = 1; i <=4; i++) {
            output.println("2 4 3 5 7 6");
        }
        output.close();

        java.util.Scanner input = new java.util.Scanner(file);

        double sum = 0;
        int i=1;

        while (input.hasNextLine()) {
            String line = input.nextLine();
            String[] values = line.split(" ");
            sum += Integer.parseInt(values[values.length - i]);
            i++;
            System.out.println(sum);
        }

    }

}
```

Answer:
6.0
13.0
18.0
21.0

2. What is the output of the following program? [1.5 marks]

```
public class Main {

     public static void main(String[] args) {
          String s1 = "Welcome";
          String s2 = s1;
          StringBuilder b1 = new StringBuilder(" CPCS203");
          StringBuilder b2 = new StringBuilder(" Exam");

          b1 = b1.replace(5, 8, "");
          s1 += b1.toString();
          b2 = b1;
          b2.insert(0, "My");

          System.out.println( s1 );
          System.out.println( s2 );
          System.out.println( b1.toString() );
          System.out.println( b2.toString() );
   }
}
```

Answer:

Welcome CPCS
Welcome
My CPCS
My CPCS

3. What is the <u>output</u> of the following program? [1.5 mark]

```
public class Test {
   int x;
   public Test(int x) {
      this.x = x;
   }
   public void myMethod(int x) {
      x = 5;
      System.out.println(this.x);
   }
   public static void main(String[] args) {
      Test t = new Test(3);
      t.myMethod(1);
   }
}
```

Answer: 3

4. What is the <u>output</u> of the following program? <u>Justify</u> your answer?

[2.0 marks]

```
public class Main {
    public static void main(String[] args) {
        Student s1;
        Student s2;
        s1.setName("Ahmed");
        s1.setGPA(5);
        s2 = s1;
        System.out.println(s1.getName() + "-" + s1.getGPA());
        System.out.println(s2.getName() + "-" + s2.getGPA());
    }
}
//---------------------------------------------------------------------
--
class Student {
    String name;
    double gpa;
    public Student() { }

    public Student(String studentName) {
        name = studentName;
    }
    public String getName() { return name; }
    public void setName(String newName) { name = newName; }
    public double getGPA() { return gpa; }
    public void setGPA(double newGPA) { gpa = newGPA; }
}
```

Answer:

Compiler error due to not constructing objects s1 and s2 using the new operator

5. As known, a method can be defined as static if it does not reference any instance data or invoke any instance method. Which methods in the following class can be defined as static? [1.0 marks]

```
public class Student {
    private int age;
    private String name;
    private static int numberOfObjects = 0;
    public int getAge() {
        return age;
    }
    public  int GetNumberOfObjects() {
        return numberOfObjects;
    }
    public void setAge(int age) {
        this.age = age;
    }
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public int lengthOfName() {
        return name.length();
    }
    public  void WelcomeMessage1(String name) {
        System.out.println("Hello " + name + "\nWelcome to the
CS course");
    }
    public  void WelcomeMessage2(String name) {
        if (name.equals(this.name)) {
            System.out.println("Hi,how are you" + name);
        }
    }
}
```

Answer:

`GetNumberOfObjects and WelcomeMessage1`

6. What is the <u>output</u> of the following program? [2.0 marks]

```
public class MidExams {
    int i;
    static int s;
    public MidExams() {
        i++;
        s++;
    }
    public static void main(String[] args) {
        MidExams f1 = new MidExams();
        MidExams f2 = new MidExams();
        System.out.println("f2.i is " + f2.i + " f1.s is " + f1.s);
        MidExams f3 = new MidExams();
        System.out.println("f3.i is " + f3.i + " f3.s is " + f3.s);

    }
}
```

Answer:

f2.i is 1 f1.s is 2

f3.i is 1 f3.s is 3

7. What is the <u>output</u> of the following program? [1.0 mark]

```
public class MidTermExam {

    public static void main(String[] args) {
        String w1 = "LEVEL";
        PalindromChecker p1 = new PalindromChecker(w1);
        String w2 = p1.getWord().append("EL").toString();
        PalindromChecker p2 = new PalindromChecker(w2);
        System.out.println(p1.IsPalindrom());
        System.out.println(p2.IsPalindrom());
    }
}

class PalindromChecker {

    private StringBuilder word;

    public PalindromChecker(String s) {
        word = new StringBuilder(s);
    }

    public StringBuilder getWord() {
        return this.word;
    }

    public boolean IsPalindrom() {
        return
word.toString().equals(word.reverse().toString());
    }
}
```

Answer:

```
false
false
```

8. What is the <u>output</u> of the following program? [1.5 marks]

```
public class Item {
    String itemName;
    double price;
    public Item(String itemName, double price) {
        this.itemName = itemName;
        this.price = price;
    }
}
public class ShoppingCart {
    Item[] items = new Item[100];

    public void addItem(Item item) {
        for(int i=0; i<items.length; i++)
            if(items[i] == null){
                items[i] = item;
                return;
            }
    }
    public void removeItem(Item item) {
        for(int i=0; i<items.length; i++)
            if(items[i] == item)
                items[i] = null;
    }
    public double getTotalCartPrice() {
        double total = 0;
        for(int i=0; i<items.length; i++)
            if(items[i] != null)
                total += items[i].price;
        return total;
    }
    public static void main(String[] args) {
        ShoppingCart cart = new ShoppingCart(100);
        cart.addItem(new Item("Memory Card", 30));
        cart.addItem(new Item("Memory Card", 30));
        cart.addItem(new Item("PC screen", 500));
        System.out.println( cart.getTotalCartPrice() );
        cart.removeItem(new Item("Memory Card", 30));
        System.out.println( cart.getTotalCartPrice() );
    }
}
```

Answer:

560

560

9. What is the <u>output</u> of the following program? [1.5 marks]

```
class Class1 {
    private int stId;
    Class1(String x) {
        this("T",2);
        System.out.println("D ");
    }
    Class1() {
        System.out.println("X ");
    }
    Class1(String y, int stId) {
        System.out.println(y+" "+stId);
    }
}
//=============================================================
class SubClass1 extends Class1
{
    SubClass1() {
        super("S",  8);
    }
    SubClass1(String y) {
        this();
    }
}
class SubClass2 extends SubClass1
{
       SubClass2() {
           System.out.println("J");
        }
}
//===============================================================
public class ContructorChanning {

    public static void main(String[] args) {
         SubClass2 st = new SubClass2();

    }

}
```

Answer:
S 8
J

10. **Given the following definitions of classes, which ones are immutable?**

[1.0 mark]

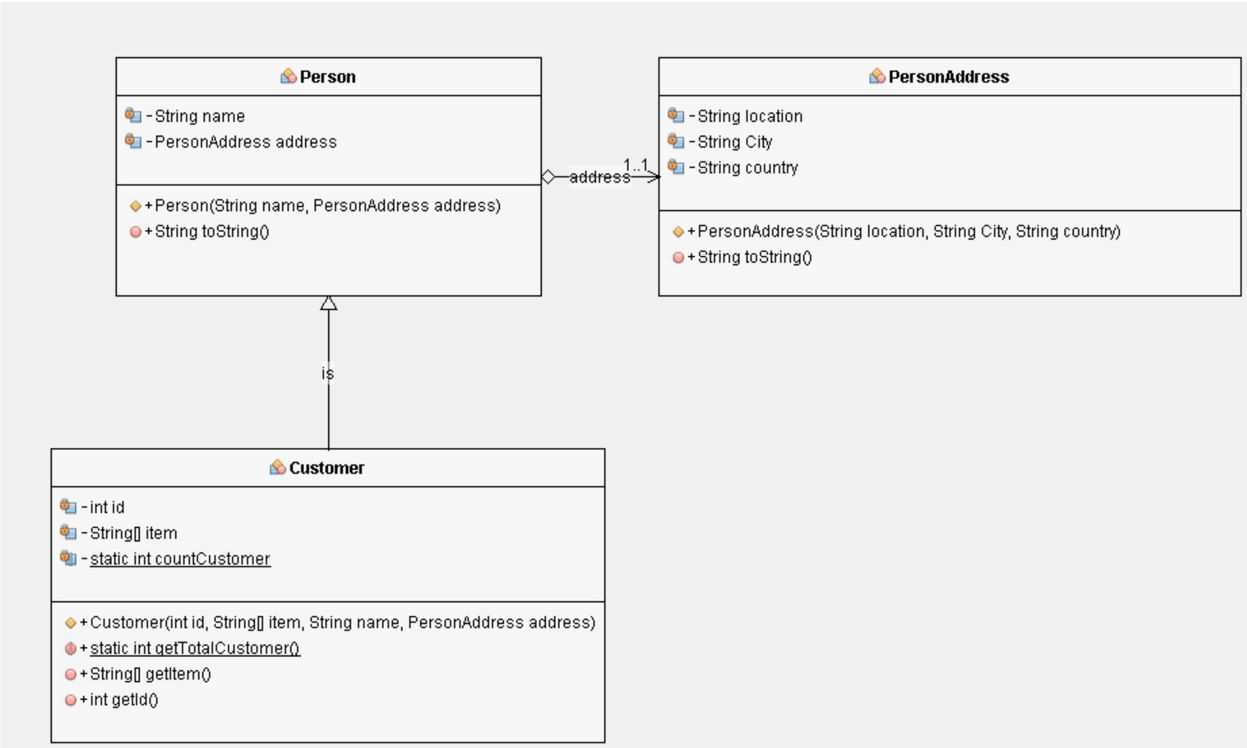
```
public class Name {
    private StringBuilder fristName;
    public Name(StringBuilder fristName) {
        this.fristName = fristName;
    }
    public StringBuilder getFristName() {
        return fristName;
    }
    public void setFristName(StringBuilder fristName) {
        this.fristName = fristName;
    }
}
//------------------------------------------------
public class Student {
    Name name;
    public Student(Name name) {
        this.name = name;
    }
    public Name getName() {
        return name;
    }

}
//------------------------------------------------
public class Teacher {
    private String name;
    public Teacher(String name) {
        this.name = name;
    }
    public String getName() {
        return name;
    }
}
```

Answer: Teacher

Part III (5.0 marks): Answer the following Question:

From the given UML class diagram, answer the following questions:



a) Translate the UML class diagram of the Customer into a Java class and make sure that the code must represent the customer class type ONLY.

[3.0 marks]

```
class Customer extends Person{
    private int id;
    private String [] item;
    private static int countCustomer;

    public Customer(int id, String[] item, String name,
PersonAddress address) {
        super(name, address);
        this.id = id;
        this.item = item;
        countCustomer++;
    }

    public static int getTotalCustomer(){
```

```java
        return countCustomer;
    }
    public String [] getItem(){
        return item;
    }
     public  int getId(){
        return id;
    }


}
```

b) The relationship between **Person** and **PersonAddress** classes is: [1.0 marks]

a) Aggregation
b) Inheritance
c) Dependency
d) There is no relationship.

c) The relationship between **Customer** and **Person** classes is: [1.0 marks]

a) Aggregation
b) Inheritance
c) Dependency
d) There is no relationship.